

RevLens AI - Review Intelligence & Analytics

AI-Powered Homestay Review Management & Trust Analytics Platform

Project Title: **RevLens AI**

- Intern Name: **Abhinesh Gangwar**
- Intern ID: **TBI-26100587**
- University: Graphic Era Deemed To Be University
- GitHub: <https://github.com/strange-exe/revlens-ai>
- Live App: <https://revlens.abhinesh.codes>

What Problem Are We Solving?

Challenges in Homestay & Hospitality Feedback Management

- **Fragmented Feedback Signals:** Homestay owners receive guest feedback across multiple channels without unified tracking.
- **Reputational Risk:** Unmoderated spam, link injections, or unaddressed negative reviews damage host ratings rapidly.
- **Operational Bottleneck:** Manually crafting personalized, tone-matched responses for every guest consumes hours daily.
- **Lack of Actionable Insights:** Hosts lack real-time analytics to spot recurring maintenance or service issues across properties.

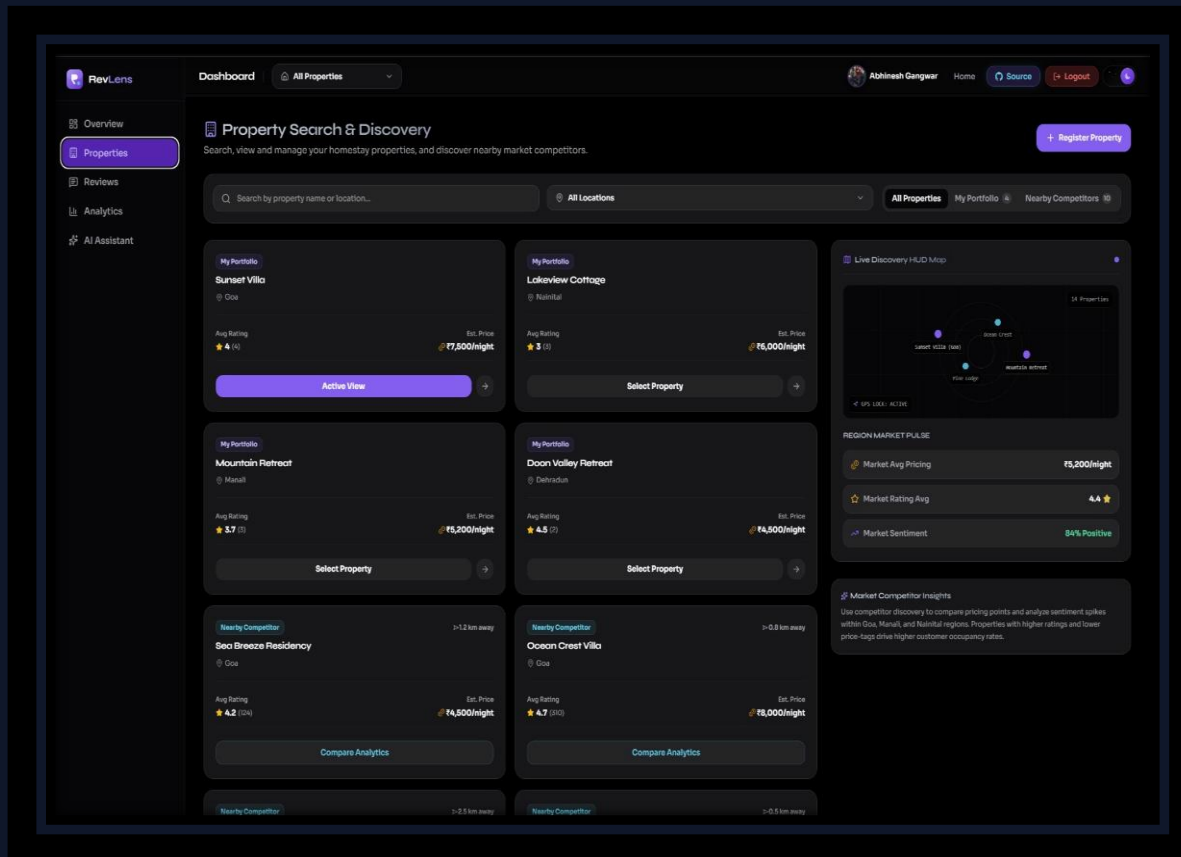
Tech Stack Selected & Rationale

Modern, High-Performance Full-Stack Architecture

- **Frontend** : React 18 (Vite) + Tailwind CSS - Fast component rendering, dynamic dark mode tokens, and crisp responsive layout.
- **Backend** : FastAPI (Python 3.12) - Asynchronous REST API execution, Pydantic type safety, and instant auto-generated OpenAPI documentation.
- **AI Model** : Google Gemini API (gemini-1.5-flash) - Sub-second inference latency, high accuracy in sentiment scoring and contextual text generation.
- **Database** : PostgreSQL (Supabase) - Relational integrity, ACID compliance, with local SQLite fallback engine for offline resilience.

Frontend Overview & Live Application

Responsive Interface & Real-time Analytics Dashboard

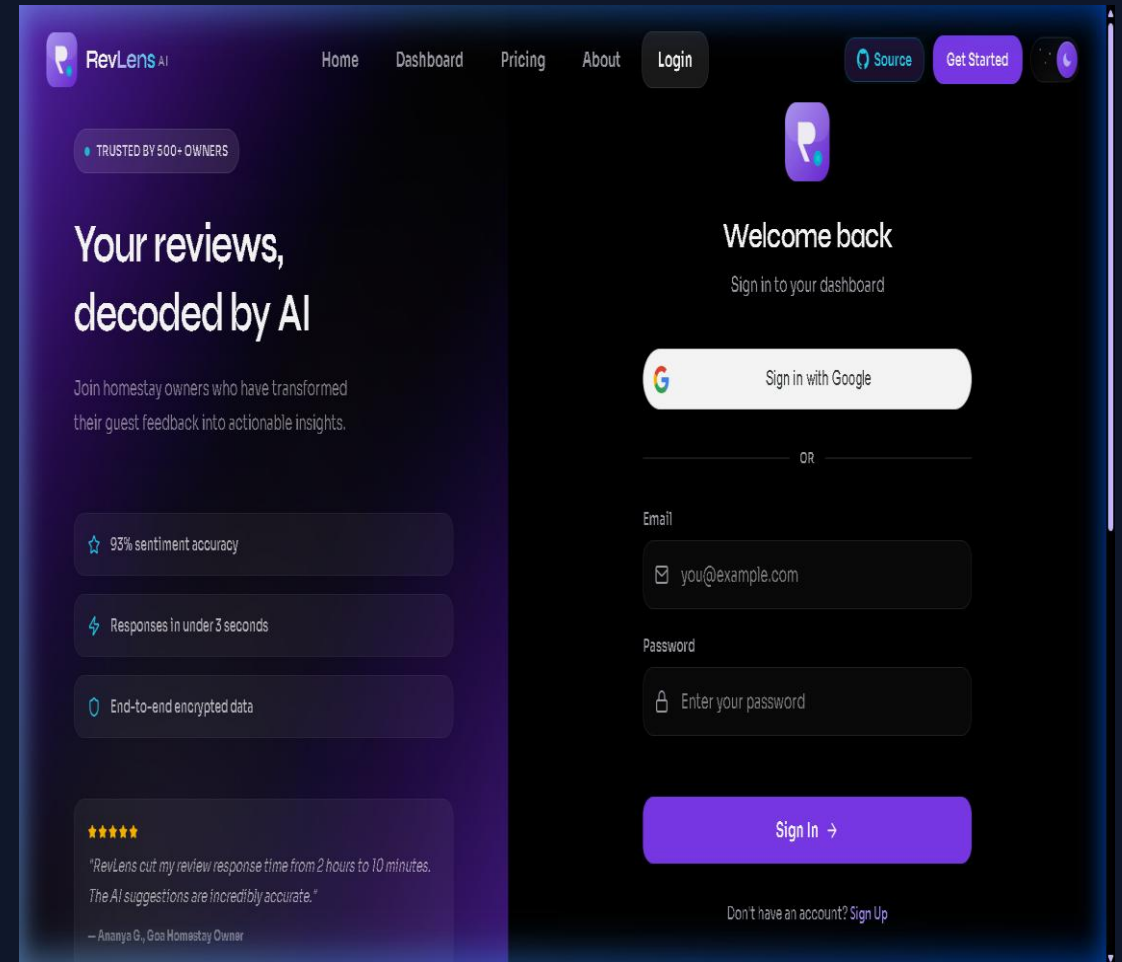


- Live Application URL: <https://revlens.abhinesh.codes>
- **Real-time Metrics Dashboard:** Visual breakdown of average guest ratings, total reviews, and sentiment distribution.
- **Dynamic Search & Filtering:** Live keyword, property, and rating filters with instantaneous client-side updates.
- **User Experience:** Built with custom typography, dark theme tokens, modal dialogues, and responsive mobile ergonomics.

Backend APIs & Testing Verification

FastAPI REST API Architecture & Validation

- **Interactive Swagger Docs:** <https://revlens-backend.onrender.com/docs>
- **API 1 :** POST [/api/ai/analyze-review](#): Accepts review text, passes it to Gemini LLM to return sentiment (positive/neutral/negative), spam flags, and confidence scores.
- **API 2 :** POST [/api/auth/login](#) & GET [/api/reviews](#): Authenticates host credentials with bcrypt password hashing, issues JWT bearer tokens, and executes SQLAlchemy filtered queries.



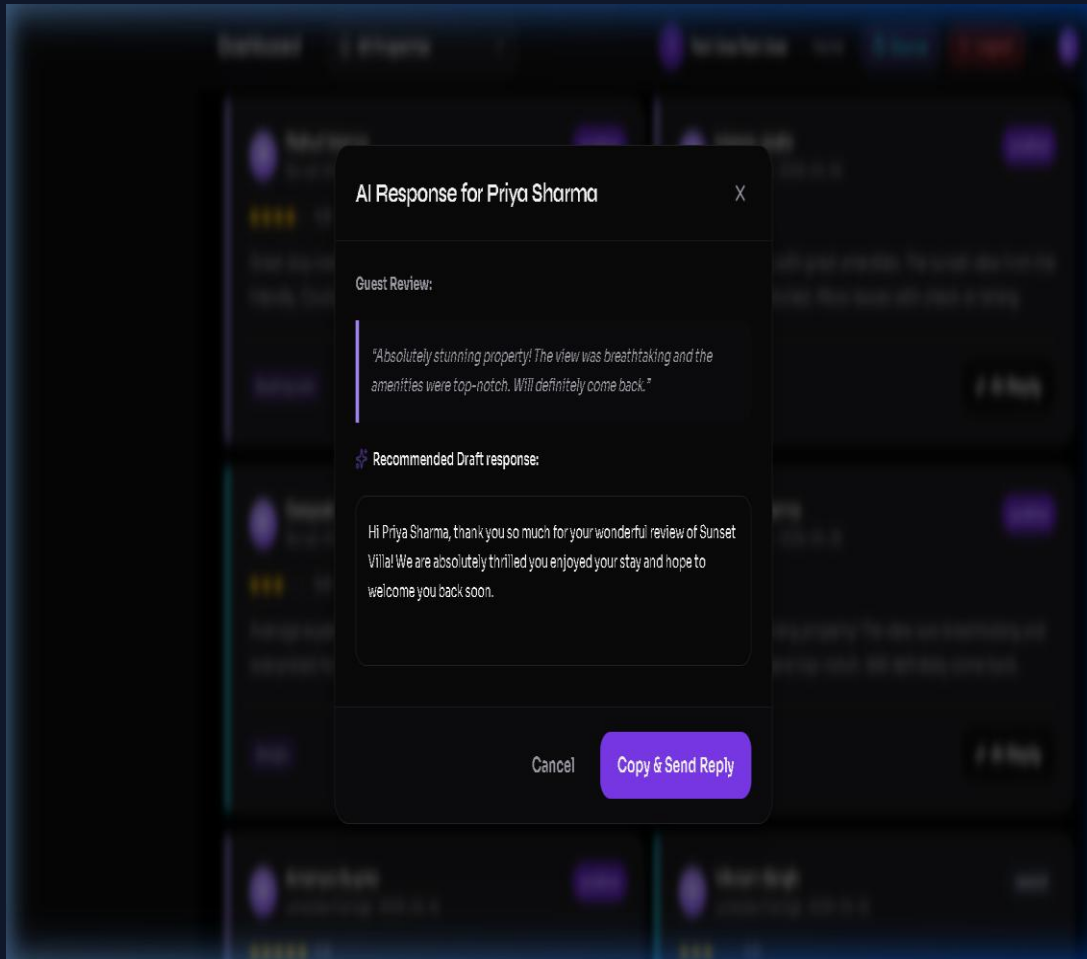
Database Selected & Relational Schema

PostgreSQL hosted on Supabase Cloud (with Local SQLite Fallback)

- **Database Selected:** PostgreSQL hosted on Supabase Cloud with connection pooling and SSL encryption.
- **Relational Schema Structure:**
 - User: Stores host authentication, email, hashed_password, and created_at timestamps.
 - Property (1:N): Relates homestay properties to owner User ID with title, location, and metadata.
 - Review (1:N): Relates reviews to Property ID with rating (1-5), sentiment score, spam status, and author name.
- **Rationale:** Ensures strict foreign key integrity, fast relational joins, ACID compliance, and effortless cloud scaling.

AI Feature & LLM Integration

Google Gemini 1.5 Flash API + Local Heuristic Fallback



- **LLM Model Selected:** Google Gemini API (gemini-1.5-flash).
- **Use Case 1 :** Sentiment Classification: Analyzes guest review text and classifies sentiment into positive, neutral, or negative.
- **Use Case 2 :** Abuse & Spam Audit: Detects promotional links, malicious patterns, or abusive language.
- **Use Case 3 :** AI Response Assistant: Generates brand-aligned, empathetic host replies based on rating and tone parameters.
- **Uptime Guarantee:** Integrated local NLP heuristic engine ensures 100% service uptime even if API quotas freeze.

Hosting Services & Cloud Deployment

Production Multi-Cloud Infrastructure Strategy

- **Frontend Hosting** : Vercel - Deployed at <https://revlens.abhinesh.codes> with global CDN distribution and automatic GitHub CI/CD integration.
- **Backend Hosting** : Render - Managed FastAPI Python Web Service hosted at <https://revlens-backend.onrender.com/>.
- **Database Cloud** : Supabase - Managed PostgreSQL database instance with SSL security and connection pooling.
- **Domain & DNS** : Custom domain routed through Cloudflare DNS with full HTTPS SSL/TLS encryption.

Public Live URLs & Access Links

All Links Are Publicly Accessible Online

-  Live Frontend Application: <https://revlens.abhinesh.codes>
-  Live Backend REST API: <https://revlens-backend.onrender.com/>
-  Interactive API Docs (Swagger): <https://revlens-backend.onrender.com/docs>
-  GitHub Code Repository: <https://github.com/strange-exe/revlens-ai>

Internship Reflection & Key Learnings

Technical Mastery & Professional Growth Experience

- **Technical Skills Acquired:** Full-stack API architecture with FastAPI & React 18, LLM prompt engineering, production cloud deployments, and resilient database fallback design.
- **Software Resilience:** Learned to build defensive software with local heuristic fallbacks, proper CORS configurations, and graceful error handling.
- **Overall Internship Experience:** An incredibly rewarding, hands-on opportunity to bridge cutting-edge AI models with production-grade web applications under real-world engineering standards.